Adaptive Multitype Contrastive Views Generation for Remote Sensing Image Semantic Segmentation

Cheng Shi, *Member, IEEE*, Peiwen Han, Minghua Zhao, Li Fang, *Member, IEEE*, Qiguang Miao, *Senior Member, IEEE*, and Chi-Man Pun, *Senior Member, IEEE*

***Abstract*— Self-supervised contrastive learning is a powerful pretraining framework for learning the invariant features from the different views of remote sensing images, therefore, the performance of contrastive learning heavily depends on the generation of views. Current view generation is primarily accomplished through different transformations, and the types and parameters of the transformations are required hand-crafted. Hence, the diversity and discriminability of generated views cannot be guaranteed. To address this, we propose a multitype views optimization method to optimize these transformations. We formulate contrastive learning as a min–max optimization problem, and transformation parameters are optimized by maximizing the contrastive loss. The optimized transformations encourage the negative sample pairs to be close and the positive sample pairs to be far apart. Different from the current adversarial view generation methods, our method can optimize both photometric transformations and geometric transformations. For remote sensing images, the geometric transformation is more critical for view generation, while the existing view optimization methods fail to achieve this. We consider the hue, saturation, brightness, contrast, and geometric rotation transformations in contrastive learning, and evaluate the optimized views on the downstream remote sensing images semantic segmentation task. Extensive experiments are carried out on the three remote sensing image segmentation datasets, including the ISPRS Potsdam dataset, the ISPRS Vaihingen dataset, and the LoveDA dataset. Results show that the learned views obtain high advantages compared to the hand-crafted views and other optimized views. The code associated with this article has been released and can be accessed at https://github.com/AAAA-CS/AMView.**

***Index Terms*— Contrastive learning, remote sensing image segmentation, view generation.**

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Cheng Shi, Peiwen Han, and Minghua Zhao are with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, Shaanxi 710048, China (e-mail: c_shi@xaut.edu.cn; 2221221154@stu.xaut.edu.cn; zhaominghua@xaut.edu.cn).

Li Fang is with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou, Fujian 362216, China (e-mail: fangli@fjirsm.ac.cn).

Qiguang Miao is with the School of Computer Science and Technology, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: qgmiao@xidian.edu.cn).

Chi-Man Pun is with the Department of Computer and Information Science, University of Macau, Macau, China (e-mail: cmpun@umac.mo).

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I. Introduction

REMOTE sensing images typically contain complex large-scale scene information, therefore, semantic segmentation is a fundamental task in remote sensing image scene understanding [1], [2]. Its primary objective is to accurately classify the scene by assigning a specific land cover category to each pixel in the image. In recent years, with the continuous development of remote sensing image semantic segmentation technology, it has been widely applied in fields such as urban planning [3], [4], environmental monitoring [5], and precision agriculture [6].

Traditional image semantic segmentation methods mainly rely on hand-designed feature extraction and rule-based classification techniques, such as scale invariant feature transform (SIFT) [7] and histograms of oriented gradient (HOG) [8]. However, these methods often have limited effectiveness in complex and diverse scenarios, making it difficult to provide robust performance. In contrast, deep learning can extract more advanced and robust semantic features. Full convolutional network (FCN) [9] is one of the pioneering methods in semantic segmentation. Subsequent methods based on FCN, such as the U-Net [10], SegNet [11], and DeepLab series [12], [13], [14], further improve and optimize the capability of semantic segmentation and almost dominate the research and application in this field. In the field of remote sensing, researchers have taken into account the specific characteristics of remote sensing images to improve the above network [15], [16]. Remote sensing images are extremely detailed and complex, leading traditional deep learning methods to encounter issues with gradient disappearance. To solve this problem, Peng et al. [17] combined dense connection and FCN to propose densely-based fully convolutional networks. In addition, the multimodal nature of remote sensing images is also a key consideration. To this end, Jin et al. [18] proposed a dense-attention-similar-fusion network that fuses information from red-green-blue (RGB) images and digital surface model (DSM) images to significantly improve the segmentation effect of the model. However, due to limited labeled samples, the interpretative ability of existing models is insufficient. Few-shot segmentation (FSS) methods have been proposed to address these challenges, particularly in remote sensing scenarios. For example, Bi et al. [19] proposed a dual-mining network (DMNet) that incorporates cross-image

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and self-mining strategies to address extreme intraclass variation. Similarly, Bi et al. [20] introduced the agent mining transformer (AgMTR) to enhance semantic clarity by adaptively aligning fine-grained semantic correlations. In addition, self-supervised learning has garnered significant attention and has been widely used in remote sensing image scene understanding [21], [22]. Self-supervised learning can learn meaningly semantic features on unlabeled samples via pretest tasks, and the downstream segmentation tasks benefit from the good representations [23], [24], [25].

In recent years, contrastive learning has emerged as a powerful self-supervised method, particularly advantageous for analyzing remote sensing images due to its ability to extract meaningful semantic features from unlabeled data. Remote sensing images are often characterized by their large scale, diversity, and multimodal information, which makes supervised learning approaches heavily reliant on extensive labeled datasets. Contrastive learning alleviates this challenge by leveraging the vast amounts of unlabeled data available, enabling the acquisition of robust feature representations through pretext tasks and facilitating their transfer to downstream applications such as semantic segmentation. Previous studies have demonstrated the efficacy of contrastive learning in enhancing model performance across various remote sensing tasks. For instance, geography-aware self-supervised learning has been employed to exploit spatially aligned images over time and utilize geographic information to design pretext tasks, significantly improving scene understanding in remote sensing [26], [27]. Zhang et al. [28] introduced a contrastive learning framework that addresses the challenges of scale and class confusion commonly encountered in land cover classification by leveraging both geographic and feature space relationships. Furthermore, the ability of contrastive learning to capture invariant representations has notably enhanced its application in remote sensing semantic segmentation, allowing models to effectively differentiate between objects in diverse and complex environments [29], [30], [31], [32]. For the complexity of remote sensing data, contrastive learning can mitigate the limitations posed by labeled samples and enhance the generalization ability of the model across diverse remote sensing scenarios.

In contrastive learning, the general idea involves maximizing the similarity between different views of the same sample while minimizing the similarity between views of different samples, therefore, one of the cores is to generate diverse views to promote the learning of discriminative features. Data augmentation is a popular way for view generation, which is primarily categorized into photometric data augmentation [33] and geometric data augmentation [34], [35]. Photometric data augmentation involves altering the pixel values of images, such as adjusting brightness and contrast, blurring, and geometric data augmentation primarily involves altering the geometric structure of images, such as rotation, scaling, cropping, and flipping. Most contrastive learning methods rely on hand-crafted data augmentation methods [36], [37], [38], which may be unsuitable for different datasets and downstream tasks. Therefore, it is crucial to generate the views adaptively.

Previous studies have investigated the adaptive view generation methods. ACL [39] and Viewmaker [40] added a perturbation to the input to produce the new view, and the perturbation is learned adversarial against contrastive loss. Kim et al. [41] designed a styler transfer network, and the augmented views are obtained by modifying the style of the input. However, these studies focus on adaptive photometric data augmentation, the adaptive geometric data augmentation in view generation has not been explored. Actually, in remote sensing semantic segmentation tasks, there exist significant differences in the direction of landcovers, such as buildings and roads, the study of geometric transformation is more crucial for generating diverse views.

In this work, we focus on both photometric data augmentation and geometric data augmentation and present an adaptive multitype contrastive view (AMView) generation method on remote sensing image semantic segmentation tasks. The adaptive views are learned in an adversarial manner. Motivated by the study in [42], we consider hue and saturation transformations in the hue, saturation, value (HSV) color space, brightness and contrast transformations in the RGB color space, as well as geometric rotation transformation. The output-augmented views are obtained by sequentially passing through the aforementioned transformations on the inputs. These transformations are first randomly initialized and then optimized along the direction of maximizing the contrastive loss. The adversarial view learning and contrastive pretraining are trained iteratively. Compared to the previous view generation methods, the proposed AMView method can explore more diverse views in a larger transformation space while improving the feature discriminative ability of the pretrained models.

To summarize, the main contributions of the article are listed below.

1) We propose a no-hand-crafted AMView generation method to learn the more diverse and discriminative views in an unsupervised manner. To the best of our knowledge, this is the first work to generate adaptive views using both photometric transformation and geometric transformation.
2) The AMViews are learned in an adversarial manner. Both photometric transformation (hue, saturation, brightness, and contrast) and geometric transformation (geometric rotation) are optimized to enhance the feature generalization and discriminative ability of pretrained models.
3) We take the experiments on three remote sensing image segmentation datasets. Compared to the hand-crafted views and current view generation methods, our method achieves excellent performance in downstream remote sensing semantic segmentation tasks.

II. Related Works

A. Contrastive Learning

As an unsupervised representation learning method, contrastive learning aims to learn generalizable and transferable representations from large amounts of unlabeled data, which can be utilized for various downstream tasks. The core idea of

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contrastive learning is to pull together the views of the same instance (positive sample pairs) while pushing apart the views of different instances (negative sample pairs). Previous studies have demonstrated the importance of negative sample selection in contrastive learning. InstDisc [43] utilized a memory bank to store negative samples, but it suffered the challenge of feature inconsistency. MoCo [44] adopted a dynamic queue to store negative samples and a momentum encoder is designed to alleviate feature inconsistency. MoCHi [45] pointed out that hard negative samples are advantageous for improving the performance of pretrained models. Therefore, a sample mixing strategy was proposed to generate hard negative samples. AdCo [46] learned the hard negative samples via a min–max optimization problem. SimCLR [47] proposed an end-to-end contrastive learning framework, which learned the discriminative features via data augmentation and nonlinear projection head. Recent works demonstrated that contrastive learning can also perform well without using negative samples. Bootstrap Your Own Latent (BYOL) [48] learned the high-quality representations by predicting the output of the target encoder from the online encoder. SiamSiam [49] proposed a simplified contrastive learning framework that eliminates the requirement of negative samples, large batches, or momentum encoders. All the methods mentioned above require hand-crafted views, while we try to learn the diverse and hard views via an adaptive way to enhance the feature discriminability of the pretrained models.

B. Good Views for Contrastive Learning

View-based contrastive learning relies on the generation of views. Some works [50], [51] have pointed out that maximizing the mutual information between views is more beneficial for feature representation. Peng et al. [52] proposed semantic-aware localization and center-suppressed sampling methods to address the issues of saliency information loss and excessive overlap areas in views. CAST [53] introduced unsupervised saliency maps to constrain the cropping process, ensuring that the augmented views contain the same objects. Zhang and Ma [54] employed an additive strategy to generate multiple views, thereby alleviating specific invariant representations introduced by data augmentation. Kim et al. [41] proposed a neural transformation network that preserves the original semantic information while generating multiple stylized views. Recently, the adversarial strategy has been introduced into view generation. ACL [39] optimized views by adding perturbations to the augmented view via the adversarial attacks method. Viewmaker [40] designed a perturbation generation network to generate views by adding perturbations to input images. However, these methods solely optimize the views through additive transformations, which restrict the diversity of the optimized views.

C. Contrastive Learning in Remote Sensing

Due to the advantages of contrastive learning, researchers in the remote sensing field have also conducted studies in various remote sensing image scenes [55]. Tile2Vec [56] is the first study to introduce contrastive learning into remote sensing image representation learning, it employs a triplet loss to push the spatially adjacent image tiles to be closer. Hou et al. [57] designed an end-to-end contrastive learning framework to pretrain the hyperspectral image classification model. Dong et al. [58] and Li et al. [59] attempted to integrate the spatial local features into contrastive learning in the semantic segmentation task. In terms of hard view generation, Li et al. [60] proposed a feature domain multiscale hard feature generation method for remote sensing image scene classification task, and the intermediate layer features are transformed to generate hard positive features and hard negative features. Chen et al. [61] designed an adversarial hard negative sample generation method for multimodal remote sensing images, and the hard-negative sample is generated by leveraging redundant information of the multimodal data. Different from the aforementioned methods, we obtain the hard positive sample and hard negative sample by optimizing the photometric transformations and geometric transformations.

III. Approach

The purpose of our method is to generate adaptive diverse views via adaptive photometric transformation and geometric transformation in an adversarial manner. For each input sample, we adaptively generate a sample pair through two groups of different photometric and geometric transformations, and feed the generated sample pair into an encoder to extract high-level discriminative features. To promote the learning of discriminative features in the encoder, we first maximize the contrastive loss to optimize the transformations, thereby generating more diverse and challenging sample pairs. Subsequently, we minimize the contrastive loss to train the encoder effectively. The overall architecture is shown in Fig. 1, and the details are shown below.

A. Contrastive Learning

In contrastive learning, the positive sample pairs are pulled together, and the negative sample pairs are pushed away. Our method is built upon a simple SimCLR structure [47]. Assume all samples in a minibatch of size N are augmented to obtain $2N$ views, and $T = \{t_1, t_2, \ldots, t_n\}$ is the set of view transformation types. For a sample x, the two augmented views are denoted as $x_i = T_i(x) = t_n^i t_{n-1}^i, \ldots, t_2^i t_1^i(x)$ and $x_j = T_j(x) = t_n^j t_{n-1}^j, \ldots, t_2^j t_1^j(x)$. The contrastive loss for the positive pairs x_i and x_j is defined as follows:

$$L(\theta, T_i(x), T_j(x)) = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} I_{[k\neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \tag{1}$$

where $z_i = f(x_i)$, f is the encoder, $\text{sim}(z_i, z_k) = (z_i^T z_k/(||z_i|| \cdot ||z_k||))$, τ is the temperature parameter, $I_{[k\neq i]}$ is an indicator function, and its value is 1 if $k \neq i$, and 0 otherwise. In traditional contrastive learning, the transformation set T is hand-crafted. In Section III-B, we design an AMView generation method to learn the transformations.

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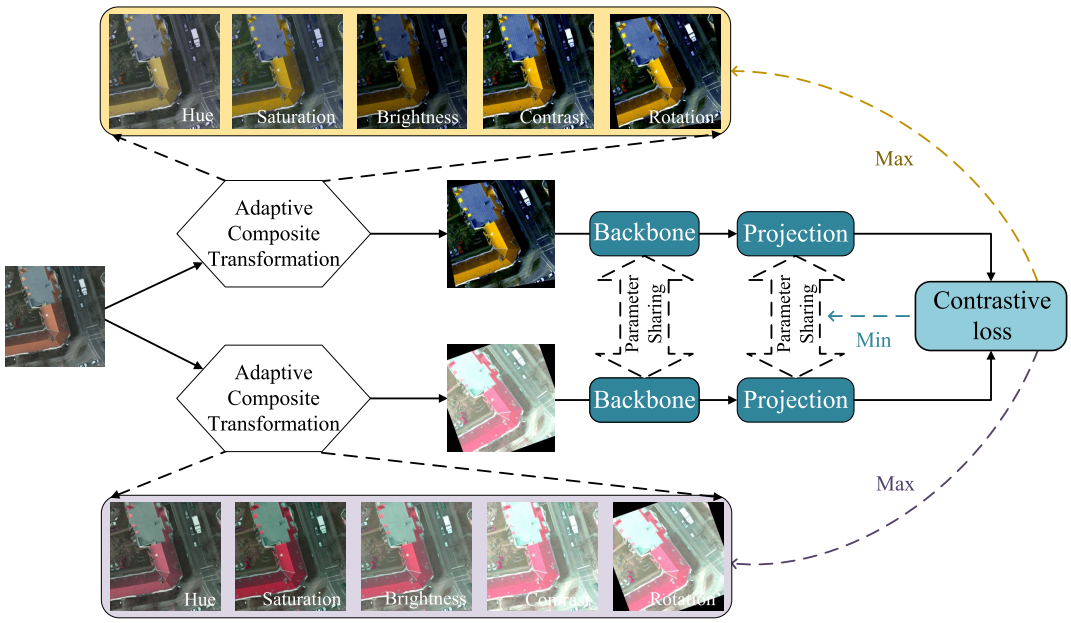

Fig. 1. Overall architecture of the proposed method. We consider five view transformations, including hue, saturation, bright, contrast, and rotation. These transformations are optimized by maximizing the contrastive loss, and the encoder with a backbone network and a projection is learned by minimizing the contrastive loss. The view transformations and encoder network are trained iteratively.

B. Adversarial Contrastive Learning

Remote sensing images contain complex landcovers, thus more diverse views are required to enhance the generalization and discriminative ability of the encoder. Obviously, the hand-crafted views do not meet the requirements of the remote sensing semantic segmentation task. Therefore, we design an adversarial contrastive learning method to adaptively learn multitype contrastive views.

The core of adversarial contrastive learning is to solve a mini–max optimization problem, which involves two key stages: inner-loop optimization and outer-loop optimization. The inner-loop optimization direction is opposite to outer-loop optimization. The objective of inner-loop optimization is the transformation parameters, aiming to generate more challenging sample pairs through optimized transformations. The outer-loop optimization utilizes the generated sample pairs to train the encoder, thereby enhancing its generalization ability. The adversarial contrastive learning is formulated as follows:

$$\min_{\theta}\left\{\max_{T_i,T_j} L(\theta, T_i(x), T_j(x))\right\} \tag{2}$$

where θ is the parameter of the encoder.

1) Inner-Loop Optimization: We first randomly initialize the transformation parameters in the two transformation sets T_i and T_j, and then optimize the parameters of transformations according to (3) and (4), respectively (the parameters θ of the encoder are fixed),

$$\tilde{T}_i = \{\tilde{t}_1^i, \tilde{t}_2^i, \ldots, \tilde{t}_n^i\} = \max_{t_i \in T_i} L(\theta, T_i(x), x) \tag{3}$$

$$\tilde{T}_j = \left\{\tilde{t}_1^j, \tilde{t}_2^j, \ldots, \tilde{t}_n^j\right\} = \max_{t_j \in T_j} L(\theta, x, T_j(x)) \tag{4}$$

where $\tilde{T}_i$ and $\tilde{T}_j$ are the optimized transformation sets. By iteratively solving (3) and (4), we can optimize the transformation parameters to increase the distance between positive pairs and keep the distance between negative pairs close. Therefore, a challenging sample pair is obtained by performing the optimized transformation sets on an input, respectively, i.e., $\tilde{x}_i = \tilde{T}_i(x) = \tilde{t}_n^i \tilde{t}_{n-1}^i, \ldots, \tilde{t}_1^i(x)$, $\tilde{x}_j = \tilde{T}_j(x) = \tilde{t}_n^j \tilde{t}_{n-1}^j, \ldots, \tilde{t}_1^j(x)$. Since the transformations do not alter the size of the samples, the augmented views have the same size as the input sample.

2) Outer-Loop Optimization: According to the optimized transformation sets $\tilde{T}_i$ and $\tilde{T}_j$, input the augmented views into the same encoder and optimize the encoder by minimizing (5) (the transformation sets T_i and T_j are fixed)

$$\min_{\theta} L\left(\theta, \tilde{T}_i(x), \tilde{T}_j(x)\right). \tag{5}$$

Next, we will describe the optimization of transformations in transformation set T_i, the optimization process for transformation set T_j follows the same procedure.

C. Optimization of Multitype View Transformations

We consider the types of view transformations to be hue, saturation, brightness, contrast transformations, and geometric rotation, which are denoted as t_h, t_s, t_b, t_c, and t_g. The same method is used to optimize (3) and (4) to obtain the augmented views, therefore, the optimization of (3) is taken as an example to illustrate the view optimization process. These transformations are optimized by performing the perturbations on them. The perturbations are randomly initialized, and optimized by solving (3) with fast gradient sign method (FGSM) [62]. The perturbation optimization for the five transformations is as follows: 1) we first optimize the hue perturbation; 2) fix the optimized hue transformation and optimize the saturation perturbation; and 3) fix the optimized hue and saturation transformations to optimize the brightness perturbation, and

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so on, until all transformations are optimized. Therefore, each transformation is optimized sequentially in an iterative manner. The details of perturbation optimization are described below.

1) Hue and Saturation: Hosseini and Poovendran [63] demonstrated that adjusting hue and saturation significantly impacts the performance of the networks, therefore, the hue and saturation transformations in the HSV color space are selected for view argumentation.

We convert the input sample x to the HSV color space, and $y_h = \mathrm{HSV}(x)$ is the hue value, with the range of $[-\pi, \pi]$. Randomly initialize the hue perturbation δ_h, and optimize the hue value according to

$$\begin{aligned} t_h(y_h) &= y_h + \delta_h \\ \tilde{\delta}_h &= \delta_h + \alpha \cdot \mathrm{sign}\left(\nabla_{\delta_h} L\left(t_g t_c t_b t_s \mathrm{HSV}^{-1}(t_h(y_h)), x\right)\right) \\ \tilde{y}_h &= \mathrm{Clip}_{[-\pi,\pi]}\left\{y_h + \tilde{\delta}_h\right\} \end{aligned} \tag{6}$$

where $t_h(y_h)$ represents a transformation of hue value, implemented by adding a hue perturbation δ_h to the hue value. $\tilde{\delta}_h$ is the optimized hue perturbation, which is optimized along the direction of maximizing the contrastive loss. α is the perturbation intensity and $\tilde{y}_h$ is the optimized hue value. $\mathrm{Clip}_{[-\pi,\pi]}$ is to clip the optimized hue value in the range of $[-\pi, \pi]$. $\tilde{x}_h = \mathrm{HSV}^{-1}(\tilde{y}_h)$ is to convert the optimized hue value into RGB color space. HSV and HSV^{-1} are the forward and inverse HSV transformations.

Saturation measures the purity of colors, with values ranging from 0 to 1. Perform HSV transformation on the sample augmented by hue transformation, $y_s = \mathrm{HSV}(\tilde{x}_h)$ is the saturation value. Randomly initialize the saturation perturbation δ_s, and optimize the saturation value

$$\begin{aligned} t_s(y_s) &= y_s \cdot \delta_s \\ \tilde{\delta}_s &= \delta_s + \alpha \cdot \mathrm{sign}(\nabla_{\delta_s} L(t_g t_c t_b \mathrm{HSV}^{-1}(t_s(y_s)), x)) \\ \tilde{y}_s &= \mathrm{Clip}_{[0,1]}\{y_s \cdot \tilde{\delta}_s\} \end{aligned} \tag{7}$$

where $\tilde{\delta}_s$ is the optimized saturation perturbation, $\tilde{x}_s = \mathrm{HSV}^{-1}(\tilde{y}_s)$ is the optimized sample after hue and saturation transformations.

2) Brightness and Contrast: The perturbations of brightness and contrast are optimized in the RGB color space. The range of brightness and contrast is from 0 to 1. Similar to (6) and (7), the perturbations of brightness and contrast are defined in (8) and (9), respectively,

$$\begin{aligned} \tilde{\delta}_b &= \delta_b + \alpha \cdot \mathrm{sign}\left(\nabla_{\delta_b} L(t_g t_c t_b(\tilde{x}_s + \delta_b), x)\right) \\ \tilde{x}_b &= \mathrm{Clip}_{[0,1]}\left\{\tilde{x}_s + \tilde{\delta}_b\right\} \end{aligned} \tag{8}$$

$$\begin{aligned} \tilde{\delta}_c &= \delta_c + \alpha \cdot \mathrm{sign}\left(\nabla_{\delta_c} L(t_g t_c(\tilde{x}_b \cdot \delta_c), x)\right) \\ \tilde{x}_c &= \mathrm{Clip}_{[0,1]}\left\{\tilde{x}_b \cdot \tilde{\delta}_c\right\} \end{aligned} \tag{9}$$

where $\tilde{\delta}_b$ and $\tilde{\delta}_c$ are the optimized brightness and contrast perturbations, and $\tilde{x}_b$ and $\tilde{x}_c$ are the transformed views.

3) Rotation: Apart from performing adaptive photometric data augmentation, we also take the adaptive geometric data augmentation into account. Specifically, we consider adaptive rotation transformation. $\tilde{x}_c$ is the input view, (c_m, c_n) represent the center position of $\tilde{x}_c$, (m', n') represent the position of (m, n) after rotating by θ degrees, which is formulated as follows:

$$\begin{bmatrix} m' \\ n' \end{bmatrix} = \begin{bmatrix} (m - c_m)\cos(\theta) - (n - c_n)\sin(\theta) + c_m \\ (m - c_m)\sin(\theta) + (n - c_n)\cos(\theta) + c_n \end{bmatrix}. \tag{10}$$

We randomly initialize an angle θ, and the optimized angle $\tilde{\theta}$ is obtained by

$$\tilde{\theta} = \theta + \alpha \cdot \mathrm{sign}(\nabla_\theta L(t_g(\tilde{x}_c, \theta), x)) \tag{11}$$

where $t_g(\tilde{x}_c, \theta)$ represents rotating the sample $\tilde{x}_c$ by θ degree. Output the augmented view $\tilde{x} = t_g(\tilde{x}_c, \tilde{\theta})$.

D. Fine-Tuning on Semantic Segmentation Task

We utilize the backbone network of the encoder for downstream semantic segmentation tasks after pretraining. The semantic segmentation network is constructed by combining the backbone network with a segmentation head. We then fine-tune the segmentation network using limited labeled samples and optimize the segmentation network using the cross-entropy loss function, as follows:

$$L_{\mathrm{seg}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} y_{ic} \cdot \log(p_{ic}) \tag{12}$$

where N represents the total number of pixels in the input image, M represents the number of classes, y_{ic} indicates whether pixel i in the ground truth label image belongs to class c, and p_{ic} is the output probability that pixel i belongs to class c.

IV. Experiments

In this section, we first provide a detailed introduction to the datasets used in our experiments. Next, we describe the experimental setup, including the implementation details of the proposed method, the selection of baseline methods for comparison, and the metrics used to evaluate the performance of the pretrained model. Finally, we analyze and discuss the experimental results through comparative and ablation studies.

A. Datasets

We evaluate the proposed method as well as other self-supervised contrastive learning methods on three remote sensing semantic segmentation datasets. These datasets are the International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam dataset, the ISPRS Vaihingen dataset, and the LoveDA dataset. The following provides a detailed description of these datasets.

1) ISPRS Potsdam: The Potsdam dataset[1] consists of 38 tiles with a spatial size of 6000 × 6000 pixels, covering six land cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. Each image has a spatial resolution of 5 cm and contains four bands: red, blue, green, and near-infrared (NIR). To generate training and evaluation samples, these tiles are cropped into nonoverlapping images with size 256 × 256 pixels, using a sliding window strategy with a step

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

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size of 256 pixels. This results in a total of 21 888 images. Among these, 13 824 images are used for pretraining, and 1% of the pretraining set is used for fine-tuning the segmentation model. The remaining images are reserved for evaluating the performance of the fine-tuned model.

2) ISPRS Vaihingen: The Vaihingen dataset[2] consists of 33 tiles with an average spatial size of 2494 × 2064 pixels and encompasses the same semantic classes as Potsdam. Each tile has a spatial resolution of 9 cm and contains three spectral bands: near-infrared, red, and green. These tiles are cropped into 33 019 images with a spatial size of 256 × 256 pixels, using a sliding window strategy with a step size of 128 pixels, allowing for the generation of more images. Out of these, 15 268 images are used for self-supervised pretraining, and 1% of the images are used for fine-tuning. The rest are used for testing the performance of the fine-tuned model.

3) LoveDA: The LoveDA dataset[3] contains 5987 high-resolution remote sensing images with the spatial size of 1024 × 1024 pixels, each comprising three spectral bands: RGB. It consists of seven classes: background, buildings, roads, water, barren, forests, and agriculture. The dataset is divided into urban and rural scenes and we focus on urban scenes. For the urban scenes, images were cropped into nonoverlapping patches of 256 × 256 pixels using a sliding window strategy with a step size of 256 pixels. This approach produces a training set of 18 496 images and a test set of 10 832 images. The training set is used for both pretraining and fine-tuning our model, while the test set is utilized to evaluate the performance of the trained model.

B. Experimental Settings

In this section, we present a comprehensive introduction to the implementation details of our proposed view generation method AMView. In addition, we describe the comparative methods utilized in the experiment and outline the evaluation metrics employed to assess its performance.

1) Implementation Details: We use the adaptive view generation method as discussed in Section III to provide diverse views for contrastive learning. Specifically, we perform multiple transformations to the input image simultaneously, including adjustments to hue, saturation, rotation angle, brightness, and contrast. Furthermore, unlike the Viewmaker method, which allows the model to learn the properties of a random function by adding a noise channel to its noise-generating network, our method achieves a similar goal by directly adding a standard normal distribution of random noise onto the original image. To control the impact of noise on the images, we reduce the noise to $1/10$ of its original value, thus introducing enough randomness while preserving the basic features of the image.

All experiments are conducted on an RTX 3090 GPU and an 11th Gen Intel Core i7-11700K CPU at 3.60 GHz. In addition, to maintain consistency with other contrastive learning methods, which commonly use a view size of 224 × 224, we applied random cropping to preprocess the data accordingly.

During the pretraining phase, ResNet18 is adopted as the backbone network and trained from scratch on each of the three remote sensing datasets. It extracts high-dimensional features, which are then mapped to a lower dimensional space via the projection head, where the contrastive loss is calculated. A detailed overview of the network architecture is provided in Table I. The optimizer used is stochastic gradient descent (SGD) with a learning rate of 0.03, weight decay of $1e^{-4}$, and momentum of 0.9. The pretraining process for the backbone network consists of 200 epochs, with a batch size set at 64. For generating the perturbations, the perturbation intensity α is set to ε_k/factor, where ε_k is calculated based on the difference between the upper and lower bounds of each perturbation. factor is a perturbation factor, which is a constant. The perturbation for each transformation (hue, saturation, rotation, brightness, and contrast) is randomly initialized within a fixed range, as shown below: hue: $[-\pi, \pi]$, saturation: $[0.7, 1.3]$, rotation: $[-10, 10]$, brightness: $[-0.2, 0.2]$, and contrast: $[0.7, 1.3]$. These ranges are chosen carefully to ensure that the transformed samples remain in their original categories. Larger perturbations are more likely to generate challenging samples, but setting excessively large ranges may cause the transformed samples to shift categories, which is undesirable for our goal of creating challenging samples while maintaining the original semantic information.

After the pretraining phase, we apply a traditional fine-tuning strategy to adjust the parameters of the DeepLabV3+ model for the downstream semantic segmentation task. During the fine-tuning process, we use the SGD optimizer with a weight decay of $5e^{-4}$ and conduct training for a total of 150 epochs with a batch size of 16. The initial learning rate is set to 0.01.

2) Comparison Methods: To assess the effectiveness of the proposed AMView method, we compare it with seven state-of-the-art contrastive learning methods, which are summarized as follows.

1) *MoCo [44]:* MoCo utilizes momentum updates and query caching for efficient negative sampling. By maintaining a dynamically updated cache of historical feature vectors, MoCo is able to sample more high-quality negatives in the generated samples.
2) *MoCo v2 [37]:* MoCo v2 introduces additional data augmentation and incorporates a projection head into its architecture, thereby enhancing contrastive learning by diversifying data samples and projecting features into a latent space to improve representation quality during training.
3) *SimCLR [47]:* SimCLR aims to maximize agreement between differently augmented views of the same image, utilizing numerous negative examples in a minibatch and applying a contrastive loss function for representation learning.
4) *BYOL [48]:* BYOL learns representations by predicting a target representation from another, using predictor and target networks to facilitate effective learning without negative samples or a contrastive loss function.

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[3]https://github.com/Junjue-Wang/LoveDA?tab=readme-ov-file

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TABLE I
DETAILED CONFIGURATION OF OUR NETWORK WITH RESNET-18 AS THE BACKBONE

	Name	Layer	Input	Output size	Ker	#Ker
Backbone	C1	conv1	224×224×3	112×112×64	7×7	64
	C2	conv2	C1	56×56×64	3×3	64
	C3	conv3	C2	28×28×128	3×3	128
	C4	conv4	C3	14×14×256	3×3	256
	C5	conv5	C4	7×7×512	3×3	512
Proj.	Pool	avg_pool	C5	1×1×512	-	-
	FC1	Linear	Pool	512	-	-
	FC2	Linear	FC1	128	-	-

TABLE II
COMPARISON OF ACCURACY WITH RANDOM INITIALIZATION AND OTHER CONTRASTIVE LEARNING METHODS ON THE POTSDAM DATASET

Method	mIoU(%)	F1(%)	mAcc(%)
Random init	39.82	54.44	53.07
MoCo [44]	45.27	59.96	58.61
MoCo v2 [37]	46.63	61.43	60.37
SimCLR [47]	47.95	62.16	61.59
BYOL [48]	46.10	60.46	59.79
SimSiam [49]	44.77	58.59	57.74
ACL(DS) [39]	47.69	61.76	61.03
Viewmaker [40]	48.05	62.10	61.16
Ours	**48.78**	**63.17**	**62.58**

5) *SimSiam [49]:* SimSiam introduces a simplified Siamese network for representation learning, eliminating the need for negative pairs, large batches, and momentum encoding.
6) *ACL [39]:* ACL enhances model generalization and robustness by introducing perturbations to generated views on top of traditional data augmentation techniques.
7) *Viewmaker [40]:* Viewmaker discards traditional data augmentation and instead introduces a noise generation network, employing adversarial learning to automatically generate views required for contrastive learning.

3) Evaluation Metric: Three widely used evaluation metrics are used for evaluating the performance of semantic segmentation tasks, including mean class accuracy (mAcc), $F1$ score, and mean intersection-over-union (mIoU).

The mAcc metric calculates the ratio of correctly classified samples to the total number of samples in each class, reflecting the overall classification performance of the model across different classes, which is calculated as follows:

$$\text{mAcc} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{13}$$

The $F1$ score combines precision and recall into a single metric that provides a comprehensive evaluation of model performance, particularly suitable for situations with imbalanced class distributions, which is calculated as follows:

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{14}$$

The mIoU metric measures the segmentation accuracy of a model by computing the ratio of intersection to union between predicted segmentation masks and the ground truth labels. It averages these ratios across all classes, providing an assessment of the algorithm's ability to accurately delineate boundaries and capture the extent of each semantic class. The mathematical representation of the mIoU metric is shown as follows:

$$\text{mIoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{15}$$

C. Experimental Results

1) Potsdam Dataset: We first evaluate our proposed approach on the ISPRS Potsdam dataset. Table II presents a comprehensive breakdown of our experimental findings from evaluating our proposed method, revealing the notable superiority of our method over the baseline SimCLR model and other contrastive learning approaches in downstream segmentation tasks, especially when dealing with limited label data. Particularly, when compared to other view-optimized-based methods, our method significantly improves the segmentation accuracy. Our method achieves remarkable improvements, boasting an approximate enhancement of 0.7% and 1% on the mIoU and mAcc metrics, respectively. These results underscore the efficacy of optimizing views across both photometric and geometric transformations, a strategy that fosters the acquisition of highly discriminative features crucial for accurate segmentation tasks.

The visualization segmentation results of SimCLR, Viewmaker, and our method are summarized in Fig. 2. The visualization results demonstrate that our method achieves an excellent performance in terms of category prediction accuracy and object edge delineation. Furthermore, to provide a more comprehensive evaluation of our method, we compute the F1 score for each class, as depicted in Table III. The F1 scores serve as a robust metric for assessing the balance between precision and recall for each class. Notably, our AMView method emerges as the frontrunner, consistently outperforming other comparative methods across most classes. This comprehensive analysis reaffirms the efficacy and superiority of our proposed approach in remote sensing semantic segmentation tasks.

Remote sensing images have the characteristics of large space size and complex content, so it is very expensive and difficult to obtain pixel-level labels. To this challenge, we propose to validate our proposed view generation method under different label counts. We randomly selected 1%, 50%, and 100% of pretrained data for fine-tuning, and the segmentation results of the Potsdam dataset are shown in Table IV. The results show that our method achieves excellent performance across different numbers of labeled samples compared to the baseline SimCLR and Viewmarker. Especially when the number of labeled samples is limited, the improvement in segmentation accuracy is more pronounced. Therefore, the pretrained model learned by our method can explore more generalizable features.

2) Vaihingen Dataset: We also validate our method on the Vaihingen dataset, and the main results of semantic

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TABLE III

$F1$ SCORE FOR EACH CATEGORY PREDICTED BY DIFFERENT METHODS ON THE POTSDAM DATASET

Method	Imp.Surf	Building	Low veg	Tree	Car	Clutter
Random init	73.31	64.31	64.71	52.37	62.41	9.52
MoCo [44]	74.44	76.00	64.20	60.68	67.50	16.94
MoCo v2 [37]	74.80	76.50	67.21	61.47	68.14	20.47
SimCLR [47]	74.22	77.79	66.57	61.91	76.09	16.40
BYOL [48]	74.03	76.18	66.40	56.03	74.30	15.79
SimSiam [49]	75.19	75.18	66.47	53.72	73.00	8.00
ACL(DS) [39]	74.21	75.62	67.57	**63.07**	75.87	14.25
Viewmaker [40]	**75.98**	78.52	**67.67**	60.53	74.72	15.20
Ours	75.79	**78.58**	65.26	62.65	**76.12**	**20.61**

TABLE IV

SEGMENTATION RESULTS AFTER FINE-TUNING WITH 1%, 50%, AND 100% OF THE LABELS ON THE POTSDAM DATASET

Method	Label Size	mIoU	F1	mAcc
SimCLR [47]	1%	47.95	62.16	61.59
Viewmaker [40]		48.05	62.10	61.16
Ours		**48.78**	**63.17**	**62.58**
SimCLR [47]	50%	59.42	72.07	71.45
Viewmaker [40]		59.47	72.67	70.79
Ours		**59.92**	**73.17**	**72.42**
SimCLR [47]	100%	65.39	77.33	75.64
Viewmaker [40]		65.62	77.25	75.71
Ours		**66.24**	**77.79**	**76.18**

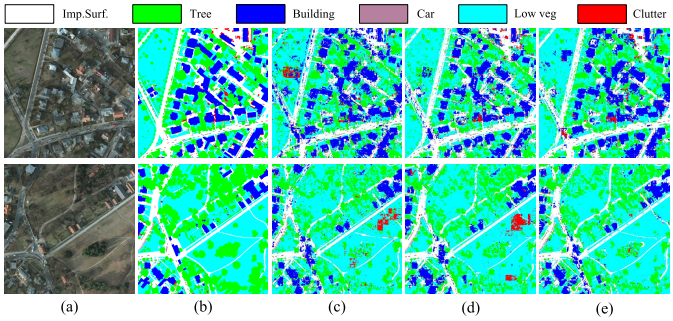

Fig. 2. Visualization segmentation results of different methods on the Potsdam dataset (image size: 6000 × 6000 for both top and bottom rows). (a) True images. (b) Ground-truth maps. (c) SimCLR. (d) Viewmaker. (e) Ours.

TABLE V

COMPARISON OF ACCURACY WITH RANDOM INITIALIZATION AND OTHER CONTRASTIVE LEARNING METHODS ON THE VAIHINGEN DATASET

Method	mIoU(%)	F1(%)	mAcc(%)
Random init	48.36	60.22	60.10
MoCo [44]	49.33	62.03	60.70
MoCo v2 [37]	49.86	62.42	61.17
SimCLR [47]	52.43	64.96	63.41
BYOL [48]	52.54	64.38	62.47
SimSiam [49]	51.97	63.12	62.90
ACL(DS) [39]	52.60	63.67	63.61
Viewmaker [40]	52.59	63.84	62.01
Ours	**53.08**	**64.98**	**63.68**

TABLE VI

$F1$ SCORE FOR EACH CATEGORY PREDICTED BY DIFFERENT METHODS ON THE VAIHINGEN DATASET

Method	Imp.Surf	Building	Low veg	Tree	Car	Clutter
Random init	80.62	82.65	68.71	82.10	45.27	1.95
MoCo [44]	83.13	83.42	67.45	81.48	49.45	7.24
MoCo v2 [37]	83.17	84.01	68.34	81.26	50.97	6.75
SimCLR [47]	83.62	85.50	71.71	82.80	**54.40**	**11.74**
BYOL [48]	85.33	86.89	72.55	83.04	52.44	6.03
SimSiam [49]	85.05	85.83	68.32	82.10	51.90	5.50
ACL(DS) [39]	84.73	86.63	70.66	82.97	50.34	6.69
Viewmaker [40]	85.34	87.78	72.60	82.81	48.83	5.68
Ours	**85.39**	**87.91**	**72.80**	**83.25**	53.18	7.36

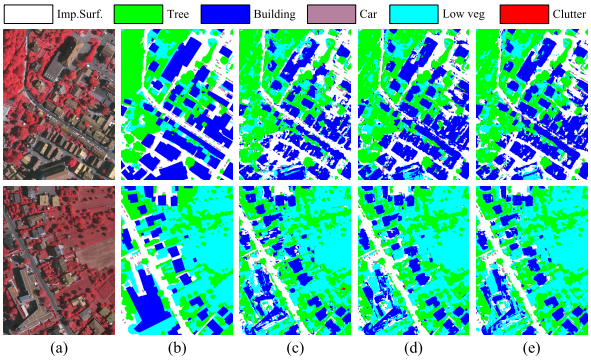

Fig. 3. Visualization segmentation results of different methods on the Vaihingen dataset (image size: 1887 × 2557 for the top row and 1903 × 2546 for the bottom row). (a) True images. (b) Ground-truth maps. (c) SimCLR. (d) Viewmaker. (e) Ours.

segmentation are presented in Table V. When using the same experimental setup, we observe similar results to those of the Potsdam dataset. Our method achieves the best performance among all comparison methods. Specifically, our method demonstrates an improvement of 0.49% in mIoU and 1.14% in F1 relative to Viewmaker, as well as a 0.27% improvement in mAcc relative to the baseline method SimCLR. To further evaluate the performance of our model, we analyze the $F1$ scores for each category predicted by different methods, as shown in Table VI. The results reveal that our method outperforms other approaches across most categories. Notably, it achieves the highest $F1$ scores in the Impervious Surface and Building categories while maintaining competitive performance in others, highlighting its robustness in handling diverse object types. Furthermore, we select several samples from the test set for visual display, as shown in Fig. 3. The results illustrate that our method exhibits higher segmentation accuracy across most scenarios, and can accurately identify and classify objects of various categories. These visual examples further validate the effectiveness of our proposed adaptive view generation method for remote sensing semantic segmentation.

3) LoveDA Dataset: Table VII details the performance of the different methods on the LoveDA dataset. We observe that the performance of Viewmaker is inferior to that of the baseline SimCLR model, specifically, the mIoU of Viewmaker

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TABLE VII

COMPARISON OF ACCURACY WITH RANDOM INITIALIZATION AND OTHER CONTRASTIVE LEARNING METHODS ON THE LOVEDA DATASET

Method	mIoU(%)	F1(%)	mAcc(%)
Random init	30.21	43.10	44.62
MoCo [44]	32.71	46.98	51.44
MoCo v2 [37]	33.60	48.28	50.07
SimCLR [47]	35.39	50.93	51.94
BYOL [48]	35.56	51.10	51.86
SimSiam [49]	33.23	47.06	51.95
ACL(DS) [39]	35.00	50.34	52.70
Viewmaker [40]	34.81	48.83	49.31
Ours	**36.04**	**51.92**	**52.82**

TABLE VIII

$F1$ SCORE FOR EACH CATEGORY PREDICTED BY DIFFERENT METHODS ON THE LOVEDA DATASET

Method	Back.	Building	Road	Water	Barren	Forest	Agri.
Random init	41.68	60.92	41.76	61.82	27.85	44.56	23.11
MoCo [44]	43.07	61.51	53.29	66.49	34.53	45.78	24.22
MoCo v2 [37]	44.10	62.04	54.91	69.28	31.93	46.24	29.47
SimCLR [47]	48.68	**66.00**	54.49	70.87	35.98	49.83	30.68
BYOL [48]	48.25	64.59	55.60	70.78	36.25	49.92	32.34
SimSiam [49]	44.65	61.72	53.19	67.36	33.03	45.96	23.54
ACL(DS) [39]	46.26	64.71	55.96	70.12	35.75	48.09	31.50
Viewmaker [40]	47.42	63.13	53.49	69.52	30.92	47.78	29.56
Ours	**49.38**	65.54	**56.67**	**71.52**	**36.79**	**50.30**	**33.22**

is lower than that of SimCLR by 0.63%. The possible reason is that the perturbations generated in Viewmaker are not stable, and only the photometric transformation is considered. In stark contrast, our approach demonstrates superior stability and robust generalization capabilities, particularly in the realm of remote sensing semantic segmentation tasks. The results underscore the critical importance of optimizing views comprehensively across both photometric and geometric domains. We also analyze the $F1$ scores for each category predicted by different methods, as shown in Table VIII. Similar to the results observed on the Potsdam and Vaihingen datasets, our method achieves the highest $F1$ scores in most categories on the LoveDA dataset. This consistency highlights the robustness and versatility of our approach across diverse datasets.

In addition, Fig. 4 shows several representative visual segmentation results, and our method performs excellently in handling details of boundaries and small regions, significantly outperforming other methods. Finally, we calculate the segmentation accuracy of all methods at different epochs, as shown in Fig. 5. Compared to other contrastive learning methods, our approach exhibits better performance in the early stages of training. Moreover, during later stages of fine-tuning, our method shows a smoother transformation curve than other methods, indicating its ability to better assist the backbone network in capturing basic features and representations.

D. Analysis and Discussion

In this section, we conduct ablation experiments on the Potsdam, Vaihingen, and LoveDA datasets to validate the

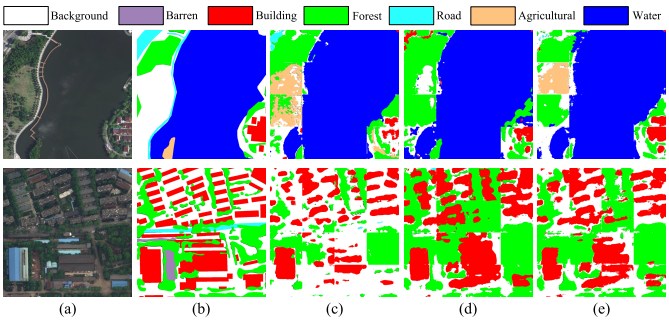

Fig. 4. Visualization segmentation results of different methods on the LoveDA dataset (image size: 1024 × 1024 for both top and bottom rows). (a) True images. (b) Ground-truth maps. (c) SimCLR. (d) Viewmaker. (e) Ours.

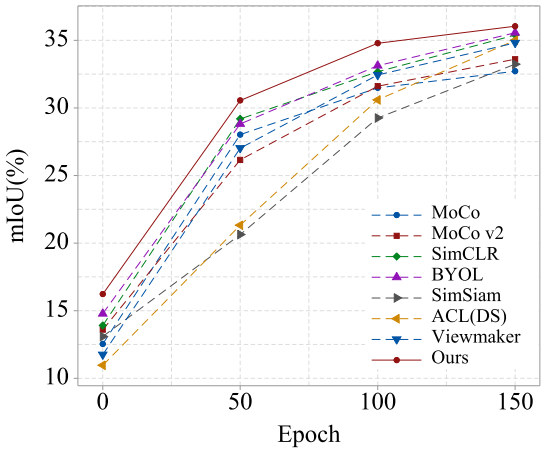

Fig. 5. Segmentation performance with different epochs in fine-tuning stage on the LoveDa dataset.

impact of each transformation on our AMView method. In addition, we investigate the perturbation intensity parameters in order to determine the optimal perturbation strength.

1) Number of Transformation Types: In the AMView method, augmented views are obtained by applying five adaptive transformations to the inputs, including hue, saturation, brightness, contrast, and rotation transformations. We investigate the effectiveness of each transformation on the pretrained model and evaluate the mIoU value on the downstream segmentation task. The segmentation accuracy is shown in Table IX. In lines 1–7, we test the segmentation accuracy with different compositions of transformations, where the composited transformations are optimized according to the method described in Section III-C. In the last line, we employ all five transformations on the inputs, and all the transformation perturbations are randomly initialized. From the observations of the lines 1 and 2, we notice that compared to the hue transformation, geometric rotation transformation is more critical for improvements in segmentation accuracy. This result is consistently observed across all three datasets.

In addition, in line 3, it can be observed that combining hue and saturation transformations leads to a 0.83% increase in mIoU on the Potsdam Dataset compared to using hue transformations alone. Therefore, it is beneficial for generating diverse views by compositing multitypes of transformations. In line 7, the composition of all optimized transformations achieves the

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TABLE IX
SEGMENTATION MIOU ACCURACY ON THE POTSDAM, VAIHINGEN, AND LOVEDA DATASETS USING VARIOUS TRANSFORMATION COMBINATIONS

NO.		Hue	Saturation	Brightness	Contrast	Rotate	**Potsdam**	**Vaihingen**	**LoveDA**
Optimized	1	✓					44.76	48.78	30.41
	2					✓	45.86	49.24	32.09
	3	✓	✓				45.59	49.67	31.18
	4	✓	✓	✓			45.30	50.32	31.11
	5	✓	✓		✓		45.95	50.37	32.29
	6	✓	✓	✓	✓		46.44	51.46	32.94
	7	✓	✓	✓	✓	✓	47.48	52.47	34.01
Random init	8	✓	✓	✓	✓	✓	46.22	51.16	31.38

TABLE X
INFLUENCE OF PERTURBATION INTENSITY ON SEGMENTATION PERFORMANCE ACROSS THE POTSDAM, VAIHINGEN, AND LOVEDA DATASETS

Perturbation factor	Potsdam			Vaihingen			LoveDA		
	mIoU	F1	mAcc	mIoU	F1	mAcc	mIoU	F1	mAcc
4	45.98	60.09	59.30	50.52	62.56	60.83	32.95	47.04	48.57
7	**47.48**	**61.99**	**61.43**	**52.47**	**63.72**	**63.17**	**34.01**	**49.38**	**49.36**
10	46.56	60.90	60.58	51.04	62.79	60.70	32.31	46.05	47.80

highest segmentation accuracy. Furthermore, in comparison to the random initialization of each transformation (line 8), the optimized transformations (line 7) exhibit an approximate 1% enhancement in the mIoU value for both the Potsdam and Vaihingen datasets. Particularly noteworthy is the LoveDA dataset, which demonstrates an even more substantial improvement with an approximate 2.6% increase in mIoU. Therefore, the proposed method is more conducive to improving the performance of downstream segmentation tasks.

2) Discussion of Perturbation Intensity: We analyze the influence of perturbation intensity on segmentation performance, the results are shown in Table X. The perturbation intensity $\alpha = \varepsilon_k/\text{factor}$, where ε_k is determined by the range of transformation perturbations, therefore, we conduct an analysis on the transformation factor factor. The transformation factors are set to 4, 7, and 10, respectively. We observe that small or large perturbation intensity all leads to a decrease in segmentation performance. Compared to a transformation factor set at 7, the segmentation accuracy in mIoU values decreases by 1.5% and 0.92% when the transformation factor is set at 4 and 10, respectively, on the Potsdam Dataset. The possible reason is that larger transformation factors fail to generate diverse views, while smaller ones may cause features of positive and negative samples indistinguishable. We aim to generate positive and negative samples that are difficult to distinguish rather than indistinguishable ones, which leads to a decrease in the performance of the pretrained model. Therefore, the optimal perturbation factor is 7. It is essential to carefully select the perturbation intensity in practical applications. The appropriate perturbation intensity can maintain a balance between the diversity of views and discriminative ability.

V. CONCLUSION

In this article, we propose an AMView method to learn the diverse and discriminative views. The augmented views are obtained through the composition of multiple types of transformations, and these transformations are optimized in an adversarial manner. The proposed multitype contrastive views generative method can optimize both photometric transformation and geometric transformation, thereby possessing stronger feature representation ability compared to other view generation methods. We evaluate the proposed method in downstream remote sensing image semantic segmentation task, and the results and analysis demonstrate the competitiveness of the proposed method.

Current view optimization methods consider the optimization of global image transformations. In our future work, we will further consider the optimization of local image transformations to better adapt to the semantic segmentation task. Moreover, since scale variations are crucial in remote sensing image segmentation, we will consider developing a more advanced architecture in the fine-tuning stage to enhance segmentation performance across different scales. These improvements will help us achieve more accurate and robust results for the challenging task of remote sensing image analysis.

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Cheng Shi (Member, IEEE) received the B.S. degree from Xi'an University of Architecture and Technology, Xi'an, Shaanxi, China, in 2009, and the M.S. and Ph.D. degrees from Xidian University, Xi'an, in 2012 and 2016, respectively.

She was a Post-Doctoral Research Fellow with the University of Macau, Macau, China, from 2016 to 2018. She is currently working as an Associate Professor with the School of Computer Science and Engineering, Xi'an University of Technology. Her research interests include deep learning, remote sensing image classification, and pattern recognition.

Peiwen Han received the B.S. degree from Xi'an University of Technology, Xi'an, Shaanxi, China, in 2022, where he is currently pursuing the M.S. degree.

His main research directions include deep learning, remote sensing image classification, and self-supervised contrastive learning.

Minghua Zhao received the Ph.D. degree in computer science from Sichuan University, Chengdu, China, in 2006.

After that, she joined Xi'an University of Technology, Xi'an, China, where she is currently a Professor with the School of Computer Science and Engineering. Her research interests include image processing, pattern recognition, and computer vision.

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Li Fang (Member, IEEE) received the bachelor's degree in land resources management and the master's degree in physical geography from China University of Geoscience, Wuhan, China, in 2007 and 2009, respectively, and the master's degree in geoinformatics and the Ph.D. degree in photogrammetry and remote sensing from the Technical University of Munich, Munich, Germany, in 2011 and 2017, respectively.

His main research interests include image processing and analysis, pattern recognition, and related remote sensing/environmental and industrial applications.

Qiguang Miao (Senior Member, IEEE) received the M.Eng. and Ph.D. degrees in computer science from Xidian University, Xi'an, China, in 2005.

He is currently a Professor with the School of Computer Science and Technology, Xidian University. His research interests include intelligent image processing and machine learning.

Chi-Man Pun (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in software engineering from the University of Macau, Taipa, China, in 1995 and 1998, respectively, and the Ph.D. degree in computer science and engineering from The Chinese University of Hong Kong, Hong Kong, in 2002.

He was the Head of the Department of Computer and Information Science from 2014 to 2019 and is currently an Associate Professor with the Faculty of Science and Technology, University of Macau. He has investigated several funded research projects as PI and has authored/co-authored more than 150 refereed scientific papers in international journals (including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, and IEEE TRANSACTIONS ON IMAGE PROCESSING) and conferences (including Association for the Advancement of Artificial Intelligence (AAAI), International Conference on Data Engineering (ICDE), Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH), and IEEE Conference on Virtual Reality and 3D User Interfaces (VR)). His research interests include image processing and pattern recognition, information security and forensics for multimedia, and adversarial machine learning.

Dr. Pun is a Professional Member of the ACM. He has served as the General Chair for the 10th and 11th International Conference Computer Graphics, Imaging and Visualization (CGIV2013 and CGIV2014) and the 13th IEEE International Conference on e-Business Engineering (ICEBE2016), the General Co-Chair for the IEEE International Conference on Visual Communications and Image Processing (VCIP2020), and the program chair/reviewer for several other international conferences and journals.

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